THANH VU

Applicant, M.S. in Computer Science (Artificial Intelligence)

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RESEARCH INTERESTS

Machine learning, probabilistic reasoning, and applied statistics for noisy real-world data, plus reliable data systems that make AI reproducible; applied research experience evaluating public data as a predictive signal.

EDUCATION

Franklin & Marshall College — Lancaster, PA

Bachelor of Arts in Computer Science | *GPA: 3.72/4.0 (Major: 3.82/4.0)* — May 2026

- Honors List (Fall 2021, 2023, 2024, 2025); Dean's List (Spring 2022)
- Computer Science: Artificial Intelligence, Machine Learning, Algorithms & Data Structures, Theory of Computation
- Mathematics & Statistics: Probability and Statistics, Statistical Modeling, Applied Regression Analysis, Information Theory

RESEARCH & APPLIED PROJECTS

Can Mail Data Predict Housing Vacancy? | *Microsoft Fabric, Power BI* — Sep 2026

- Investigated whether quarterly USPS vacancy counts can stand in for the annual Census ACS vacancy rate, building a reproducible pipeline over 6 counties and 15 years (**84 county-year observations**).
- Evaluated the hypothesis three ways (correlation, directional accuracy, bias), finding a pooled **r = 0.77** that ranged from −0.01 to 0.72 by county and only **57.7%** directional agreement, so USPS data is a county-dependent early signal.
- Isolated a 2012 USPS classification change as a data artifact and fixed two silent ingestion bugs, preventing a false trend.

Grocery Transaction Data Warehouse | *SQL Server, T-SQL* — Jul 2025 – Aug 2025

- Built a Medallion-architecture warehouse over **131M+ transaction lines** with stored-procedure ETL, a star-schema Gold layer, and 21 automated data-quality checks; query tuning cut processing time by **25%**.

Medical Cost Regression | *R, ggplot2, R Markdown* — Nov 2021

- Compared five nested linear models, including a sex × age interaction, on **1,338 insurance records** with residual diagnostics, showing age is a significant predictor while sex adds little once age is controlled.

AI, Systems & Theory Coursework | *Python, C++, RISC-V* — 2024 – 2025

- Implemented uninformed and informed search, constraint satisfaction, logical inference, and Bayesian networks in Python, and solved a Connect-4 board puzzle with state-space search.
- Engineered features and explored real-world datasets (Steam games; 13.6K Melbourne housing listings) in pandas for Machine Learning; designed a CPU in Logisim, wrote recursive RISC-V assembly, coded DFA/NFAs, and benchmarked a red-black tree.

EXPERIENCE

Data Analyst Fellow | EDC Lancaster County (Center for Regional Analysis), *Lancaster, PA* — Aug 2026 – Present

- Automated 25+ federal economic data series (BLS, BEA, Census, USPS) into Power BI and Google Apps Script dashboards with daily refresh, saving **15+ hours per month**.
- Embedded a **Gemini LLM assistant** in a Google Apps Script dashboard that summarizes trends and flags anomalies, letting 6+ non-technical staff explore and update data without code.

Data Science Intern | KCC Capital Partners, *Sheridan, WY* — Jan 2026 – May 2026

- Applied statistical and machine-learning models to data on **100+ mining and precious-metal companies** to shortlist 100+ potential buyers for active mandates, and automated weekly reports that save 5+ hours per week.

Data Engineer Intern | KPIM Joint Stock Company, *Hanoi, Vietnam* — Jan 2021 – Apr 2021

- Engineered Python and SQL Server ETL pipelines that automated legacy client data workflows, reducing processing time by **30%**.

Technology Support Assistant | Quantitative Science Center, Franklin & Marshall College — Sep 2021 – Sep 2022

- Resolved **50+ technical issues per week** for students and faculty, maintaining **99% classroom uptime**.

SKILLS & CERTIFICATIONS

Programming: Python (Pandas, NumPy, Matplotlib, PySpark), SQL (T-SQL, Spark SQL), R, C++, Java, C#, RISC-V Assembly, JavaScript
ML & Statistics: Regression, statistical inference, probabilistic reasoning (Bayesian networks), machine learning, model evaluation
Data Systems: Microsoft Fabric (Lakehouse, Dataflow Gen2, Notebooks), SQL Server, Power BI, ETL, data modeling, Git
Certifications: Microsoft Certified: Fabric Data Engineer Associate (DP-700), Oct 2026; HackerRank SQL (Advanced), Sep 2025

LEADERSHIP

Technology Committee | Diplomatic Congress, Franklin & Marshall College — Aug 2021 – Aug 2022

- Cut classroom technology costs by **18%** by shaping campus technology policy with 10+ committee members.